NET MANAGER SECURE USER GUIDE

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1 Introduction

The Net Manger Secure (net_mgrS) application is a wrapper for Net Manager (net_mgr) that adds encryption, integrity, and authentication to net_mgr operation via the use of secure associations. The creation and use of secure associations requires a shared secret, which is protected by and uses a keystore, a Keysafe, or High Strength KeySafe (HSKS). In addition to the creation of secure associations, net_mgrS will automatically perform other secure association management functions, such as renewing expired associations.

Net Manger Secure automates the secure association processes, giving users with access to KeyKeeps or HSM configurations and easy, secure access to net_mgr functions.

Net ManagerS provides:

- A relatively long-lived secure association between service points, typically lasting for 5 days.
- Renegotiation of associations at will, for either the client or the server.
- Strong authentication of the client to the server, and the server to the client, by way of certificate chain validation.
- Transmission of required certificates in-line.

  Authorization choices of the client are determined by permissions encoded into the client's certificate, thereby allowing large-scale authorization decisions without substantial overhead or configuration (no authorization server needs to be contacted to validate clients.)
- Created associations that begin with a pair of key pairs, one for the server (typically the communications module) and one for the client (typically a back office system).
- Use of ephemeral keys (short-lived operational keys that are derived from the keys in the certificate chain and may be rotated at will).
- Use of high-speed symmetric cryptography algorithms used once an association is formed from the public/private key pairs.
- (Optional) use of CryptKeeper for crytographic tasks. (If you use HSKS, CryptKeepr is required.)
- Primitive cipher negotiation (AES-256-CFB is the only supported mode in this release).
- Use of HMAC-SHA256 for authenticating that packets belong to the correct association.

Note: If you have turned your legacy port off, net_mgrS is required.

For Further Information

For more information about security in the Silver Spring Networks environment, refer to:

- *Silver Spring Networks Security Overview*
- *Enabling Link Layer Security*
- *Critical Operations Protector (COP) User Guide*
- KeySafe and C*ritical Operations Protector (COP) Administration and Monitoring Guide*
- *Certificate Authority Tools User Guide for Operators*

2 Net Manager Secure Commands

Since net_mgrS provides secure functionality to net_mgr, net_mgr has to be available. The specific version of net_mgr that must be available is based on the following logic:

1. For PATH environment variable: net_mgrS will use whatever net_mgr comes up first on the specified path.
2. For NET_MGR environment variable: net_mgrS will use the net_mgr instance specified in this environment variable over that specified in the PATH variable.
3. For NMS_NET_MGR environment variable: Same functionality as NET_MGR, but NMS_NET_MGR will be used if both are defined.
4. For --net_mgr=xxxx or --nm=: When passed as a long argument, net_mgrS will use that above any other. See Table 2 for a list of long arguments.

Configuring Options

There are two methods for setting certain options:

- Environment variables shown in Table 1
- Command line options shown in Table 2.

Command line options will override environment variable settings if you are using both methods.

Net Manager Secure Environment Variables

The net_mgrS command options (listed in Table 2) always override environment variables.

Table 1. net_MgrS environment variables

Environment Variables	Usage
NMS_ASSOCSTORE	Use the specified file for the association store.
NMS_KEYKEEP	Use the specified file for the keykeep.store.
NMS_NET_MGR	Specifies the binary to use for net_mgr invocations.
NMS_C1219_OP	Specifies the binary to use for c1219_op invocations.
CS2_PKCS11_INI	Specifies the cs2_pkcs11.ini file path that contains Utimaco HSM locations.
NMS_HSM_SLOT	Specifies the HSM slot number that will be used (`--slot`).
NMS_HSM_AUTH_FIL	Specifies the file that contains the HSM slot authentication parameters (--authfile)

The following are examples of exporting environment variables:

```
$ export NMS_NET_MGR=/home/jsmith/usr/bin/net_mgr_2.9
$ export
NMS_CK_URL=https://discofever.rwc.silverspringnet.com:9443/cryptkeeper
$ export NMS_CK_USER=root
$ export NMS_CK_PASSWORD=xxxxx
```

Net Manager Secure Options

Table 2 lists the supported options for the `net_mgrS` command.

Table 2. net_mgrS command options

nett_mgrS Options	Description
`--`	Specifies that what follows is passed directly to net_mgr. Used when long options must be passed to net_mgr.
`--0`	Shortcut for --here
`--assocstore=<path>`	Set the location of the association store. Overrides `NMS_ASSOCSTORE` command.
`--auth=<pin>`	For HSM: Add an authorization code.
`--a`	Shortcut for `--auth`.
--authfile=<file>	For HSM: Read authorization codes from a file.
--assockey=<key>	Key used to look up an association (-d otherwise)
--Ask	For HSM: Prompt for an authentication code.
`--A`	Shortcut for --Ask.
--blobfile=<file>	Set the location of the blobfile.
--blobSkey=<file>	Set the location of the blobfile Skey.
--bulkassoc=<file>	Specifies bulk association mode. net_mgrS will ensure an association with every destination in <file>. Destinations are specified exactly as with -d.
--c1219_op=<arg>	Use the `c1219_op` invocation <arg>.
--CS2_PKCS11_INI=<arg>	Use `arg` as CS2_PKCS11_INI file.
--C=<arg>	Shortcut for --CS2_PKCS11_INI.
--debug	Debug level logging output.
--force	Force a new association.
--help	Displays this list of support options for the net_mgrS command.
--here	Insert the crypto arguments here in the list of net_mgr arguments. Applies to manual operations.
--HSM_PROPS=<file>	Load HSM properties from specified properties file.
--H=<file>	HSM_PROPS Note that properties are NICNAC names. Specifically ENTITY_SLOT and PIN are used. See `--prefix`.
--keykeep=<file>	Set the location of the keykeep.store file <default ./>. (Overrides NMS_KEYKEEP.)

Table 2. net_mgrS command options

nett_mgrS Options	Description
--manual	Locate the crypto arguments manually (used in case net_mgr has arguments that net_mgrS does not know about.)
--nm=<arg>	Shortcut for --net_mgr.
--nms_selftest	Run a self-test routine.
--net_mgr=<arg>	Use `net_mgr` invocation <arg>. (Overrides NMS_NET_MGR.)
--no_rm	Do not delete temporary files at shutdown.
--no_sa	Do not associate, just exit if we cannot continue without one.
--N=<file>	Shortcut for `--authfile`.
--pk11lib=<arg>	Provide a path or search term for the pkcs11 driver library, for example: `libcs2_pkcs11.so`.
--prefix	Prefix to use for HSM properties, for example, `SSN_SSNGMR`.
--pushnm	Push NM chain of certificates to the device.
--pushop	Push operator certificate to the device.
--pushpermit	Push permit chain of certificates to the device.
--quiet	Suppress association output.
--slot	HSM slot to log into.
--s	Shortcut for --slot.
--trace	Specifies maximum logging output.
--verbose	Specifies verbose (debug) level logging output.
--version	Displays the version.

Debugging Commands

Net Manager Secure takes a `--debug` flag that shows what calls are made to net_mgr. You can use the **`net_mgrS --debug`** command to emit the binary found for net_mgr.

The following is sample output from the `--debug` option:

```
$ ./net_mgrS --debug -d mirnezami echo 1
DEBUG net_mgrS.c:1108  ** Addressing device mirnezami
DEBUG net_mgrS.c:1274  DB Association ID is: 40536
DEBUG net_mgrS.c:1275  DB Shared secret is:
b77821ab021f53bc025e7e5025831da024dbe45e339372832b735be2452a2955
DEBUG net_mgrS.c:1276  DB seqnum is: 5
DEBUG net_mgrS.c:1310  net_mgr -e 06 -a 03 -A 40536 -k
b77821ab021f53bc025e7e5025831da024dbe45e339372832b735be2452a2955 -c
5 -d mirnezami echo 1
Ok
Echo result (resp len = 1) :
```

You can also specify `--trace` to output a more detailed level of debugging output.

Note: The --trace option generates a high volume of output.

3 Net Manager Secure and CryptKeeper

Net Manager Secure uses CryptKeeper for cryptography.

The CryptKeeper command line options are listed in Table 3.

Table 3. CryptKeeper command options

CryptKeeper Command Line Options	Export Syntax	Description	Usage
`--ck_url`	NMS_CK_URL	Specifies the URL of the CryptKeeper server.	Required
`--ck_selector`	NMS_CK_SELECTOR	Specifies a string which identifies the nmentity chain.	Optional
`--ck_user`	NMS_CK_USER	CryptKeeper user name for HTTP basic authentication.	Required
`--ck_password`	NMS_CK_PASSWORD	CryptKeeper password for HTTP basic authentication.	Required
`--ck_no_ssl_verify`		Disable verification of CryptKeeper SSL server certificate.	Optional

The following is an example of using CryptKeeper command line options with the net_mgrS command:

```
./net_mgrS --net_mgr=./net_mgr
--ck_url=https://discofever.rwc.silverspringnet.com:9443/cryptkeeper
--ck_selector=STANDARD --ck_user=root --ck_password=xxxxxx
--ck_no_ssl_verify -d fd71:4af4:530c:117:213:5001:012a:db0b echo 1
```

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